Product Requirements Document

Adaptive AI–NWP Weather Blending Platform

Smart India Hackathon — Problem Statement SIH26081

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Location / MVP city: Bhopal, Madhya Pradesh

Status: Draft — for team review

1. Overview

This document defines the requirements for a hybrid AI–NWP forecast blending platform, built for SIH26081. The core deliverable is a genuinely working adaptive blending engine that combines a physical NWP model, an AI/ML model, and an ensemble model into a single optimized forecast for rainfall, temperature, and wind, with weights that adapt by time period based on historical model skill. A supporting application layer (citizen and farmer-facing features) demonstrates the engine's output in a usable product context.

2. Problem Statement

Physical NWP models, ensemble forecasts, and AI/ML weather models each perform differently depending on region, season, lead time, and weather regime — no single source is reliably best across all conditions. The challenge is to build a framework that assigns adaptive weights to each forecast source based on historical skill, and produces a blended forecast that measurably outperforms any individual source, along with transparency into which source is being trusted and why.

3. Goals & Success Criteria

Primary goal (must succeed)

- Ship a working blending engine for Bhopal that computes adaptive, skill-based weights across three forecast sources and produces a blended output for rainfall, temperature, and wind.
- Demonstrate, with backtested numbers, that the blend outperforms every individual model source on held-out historical data.

Secondary goals

- Present the blend's weighting and confidence in a form a non-technical user can understand and act on (trust map, confidence label, explainable alert card).
- Show a working farmer-advisory slice built directly on the blended output, proving the engine is usable in a real decision-support context.

Tertiary goal

- Present the broader platform vision (marine, disaster-management, travel, voice) as clearly labeled “Coming Soon” options, to show extensibility without overclaiming what is built.

4. Scope

In scope for MVP

- One city (Bhopal) as the region for the working blending engine.

- Three variables: rainfall, temperature, wind.
- Three forecast sources: one physical NWP model, one AI/ML model, one ensemble model.
- Time-based adaptive weighting: each calendar month split into three 10-day periods, each with its own weight set per variable.
- Output as both a single blended value and a range with a High/Medium/Low confidence label.

Explicitly out of scope for MVP

- Additional cities/regions beyond the single-city proof-of-concept (documented as extensible, not built).
- Lead-time stratification finer than the current MVP horizon (7-day and 3-day period granularity are stated as post-MVP stretch goals).
- Marine, disaster-management, travel, and voice-assistant modules — shown in-product as “Coming Soon.”

5. Target Users

- **Normal user —** A general citizen in or near Bhopal checking daily/weekly weather.
- **Farmer —** A farmer in a Bhopal-area Panchayat needing sowing, irrigation, and spray-timing guidance.
- **SIH evaluator —** A judge/evaluator assessing whether the blending methodology is real, measurable, and technically sound.

6. Data Sources

All data is sourced through Open-Meteo, a free, no-API-key weather data provider, so the pipeline has no paid or access-restricted dependency.

- **Physical NWP —** NOAA GFS or DWD ICON, via Open-Meteo's Forecast API.
- **AI/ML model —** ECMWF AIFS (ECMWF's operational machine-learning model), via Open-Meteo's ECMWF Forecast API — a genuine AI/ML source with no custom model training required.
- **Ensemble model —** ECMWF IFS ENS / GFS ENS / ICON EPS, via Open-Meteo's Ensemble API.
- **Historical forecasts —** Open-Meteo's Historical Forecast archive (from 2021) and Previous Runs API (fixed 1–7 day lead-time offsets, from January 2024).
- **Ground truth —** ERA5 reanalysis (from 1940, gap-free), used to score each model's historical accuracy.
- **Supplementary —** Optional cross-check of ERA5 against IMD station observations for Bhopal via data.gov.in, for local credibility — non-blocking, added after the core pipeline works.

7. Blending Methodology

Time buckets

Each calendar month is split into three 10-day periods. Each period gets its own weight set, computed independently, so the engine adapts within a month rather than only across seasons. Splitting into 7-day and then 3-day periods is a stated post-MVP refinement using the same method at finer granularity.

Skill scoring

- Temperature and wind: RMSE (root-mean-square error) of each model's historical forecast against ERA5 ground truth, computed per 10-day period.
- Rainfall: a categorical skill score (CSI or ETS) against observed rainfall thresholds, computed per period, since rainfall skill is better judged by hit/miss/false-alarm behavior than raw error.

Weight computation

For each period and each variable, each model's error is inverted (lower error = higher skill) and normalized across the three models so the weights sum to 100%:

weight_i = (1 / error_i) ÷ Σ (1 / error_j) for all models j

This produces a separate weight table per variable, per 10-day period — e.g., early-February rainfall may weight Model A at 55%, Model B at 35%, Model C at 10%, while early-February temperature weights the same three models differently.

Validation

- Weights are computed using several years of historical data pooled per period, to avoid instability from any single unusual year.
- The blend is tested on a held-out year the weight computation never saw, so the reported skill improvement is not circular.
- Result is surfaced as the Forecast Accountability Score: blended-forecast skill vs. each individual model's skill, on the same held-out data.

Output format

- A single blended value per variable (e.g., “38 mm rainfall”, “31°C”).
- A range with a confidence label (e.g., “35–60 mm, Medium confidence”), where confidence is derived from model agreement, ensemble spread, and the period's historical accuracy.

8. Functional Requirements

Tier 1 — Core Engine (must be fully working)

ID	Requirement	Description	Priority
FR-1	Blending engine	Combine NWP, AI/ML, and ensemble forecasts into one output using period-specific, variable-specific weights.	Must
FR-2	“Which model to trust?” view	Show the current weight assigned to each source for the active period/variable.	Must
FR-3	Confidence & Uncertainty Engine	Assign High/Medium/Low confidence to every forecast from model agreement, ensemble spread, and historical accuracy.	Must
FR-4	Forecast Accountability Score	Backtest and display blended vs. individual-model skill on held-out data.	Must
FR-5	“What changed?” delta view	Show how a forecast has shifted since the previous update.	Should

Tier 2 — Demonstration Layer (working prototype)

ID	Requirement	Description	Priority
FR-6	Location search	Find weather by city, village, district, or map pin.	Must
FR-7	Current & forecast display	Show current conditions plus hourly/24h/72h/5-day outlook from the blended engine.	Must
FR-8	Multiple saved locations	Save home/work/village/travel/family locations.	Should
FR-9	Location-specific notifications	Alert per saved location individually, reusing the same	Should

ID	Requirement	Description	Priority
		threshold logic.	
FR-10	Threshold-based push notification	One or two configurable alert types (rain %, heatwave threshold).	Should
FR-11	Language toggle	Hindi / English / Hinglish.	Could
FR-12	Explainable alert card	Show what/when/risk/action for any serious alert, surfacing FR-1–FR-3 output in plain language.	Must
FR-13	Farmer profile & Panchayat outlook	Village/crop/crop-stage profile with a localized slice of the blended forecast.	Must
FR-14	Crop-stage advisory & protection alert	Sowing/irrigation/spray suitability and crop-protection alerts from the blended forecast.	Must

Tier 3 — Coming Soon (visible in UI, not built)

ID	Requirement	Description	Priority
FR-15	Marine module	Wave/swell/tide data and a Go/Caution/No-Go trip card.	Won't (MVP)
FR-16	Disaster-management dashboard	Multi-hazard mapping and resource planning for authorities.	Won't (MVP)
FR-17	Travel Weather Guardian	Route-aware risk monitoring for a planned journey.	Won't (MVP)
FR-18	Voice-first advisory agent	Local-language voice Q&A with spoken confidence.	Won't (MVP)
FR-19	AQI / fire-hotspot / satellite / radar layers	Additional environmental overlays.	Won't (MVP)
FR-20	Panchayat MicroCast downscaling	Terrain/land-use-based downscaling to Panchayat resolution.	Won't (MVP)
FR-21	Verified crowdsourced feedback	User-submitted reports cross-checked against station/radar data.	Won't (MVP)
FR-22	Farmer voice/feedback/station display	Polish layered on top of FR-13/FR-14.	Won't (MVP)

9. Technical Architecture

- **Data ingestion —** Python (requests/httpx + pandas) pulling forecast, historical-forecast, and ERA5 data from Open-Meteo for Bhopal.
- **Storage —** SQLite for the MVP (single city, moderate data volume); migrate to Postgres if scope expands to more cities.
- **Weight computation —** A pandas-based module computing RMSE/CSI per model, per 10-day period, per variable, then normalizing into weight tables (offline, scheduled recomputation).
- **Serving layer —** FastAPI exposing the current blended forecast, weight table, and confidence/accountability data as JSON.
- **Dashboard —** Streamlit for the fastest path to a judge-facing demo (trust map, confidence bars, accountability chart); React only if frontend bandwidth allows a more polished build.

10. Success Metrics

- Blended forecast shows a measurable skill improvement (lower RMSE for temperature/wind, higher CSI/ETS for rainfall) over each individual model on held-out data.
- Weight tables visibly differ across at least two 10-day periods with different weather character (e.g., a monsoon period vs. a winter period), proving the adaptation is real and not a fixed average.
- Confidence labels correlate with actual forecast error — periods labeled “Low confidence” should show measurably higher error than periods labeled “High confidence” in backtesting.
- A judge can trace one full path end-to-end in the demo: blended forecast → confidence label → explainable alert → farmer recommendation.

11. Milestones

- **Phase 1 —** Set up Open-Meteo pipeline; pull historical forecasts + ERA5 for Bhopal; confirm data availability across the needed years.
- **Phase 2 —** Compute per-period, per-variable RMSE/CSI for each model; generate normalized weight tables; validate on held-out year.
- **Phase 3 —** Build FastAPI serving layer exposing blended forecast, weights, and confidence.
- **Phase 4 —** Build Streamlit/React dashboard: trust map, confidence display, accountability score, explainable alert card.
- **Phase 5 —** Build farmer profile + advisory slice on top of the same API.
- **Phase 6 —** End-to-end test, tighten the demo narrative, prepare presentation.

12. Risks & Assumptions

- Historical forecast archive depth varies by model on Open-Meteo (e.g., ECMWF IFS history starts March 2024; other models from September 2025) — the held-out validation window must be chosen within the years each model actually has archived data.
- 10-day period buckets may have a small sample size in a single year; weights should pool multiple years per period to stay stable.
- Rainfall skill scoring (CSI/ETS) requires choosing sensible rainfall thresholds for “heavy rain” in Bhopal — to be confirmed during Phase 2.
- Tier 2 farmer advisory logic (sowing/spray/irrigation suitability) needs a simple rule-set to be defined — to be drafted alongside the crop-stage feature, not left implicit.

13. Open Questions

- Exact rainfall thresholds to treat as “heavy rain” for CSI/ETS scoring in Bhopal.
- Which specific NWP/ensemble model pair (e.g., GFS vs. ICON) performs better as the MVP's physical-model source — to confirm empirically in Phase 1.
- Team role assignment across the six phases above.